

Course: Networked Control

Installation of Yalmip

Prof. Marcello Farina

Dipartimento di Elettronica, Informazione e Bioingegneria (DEIB)

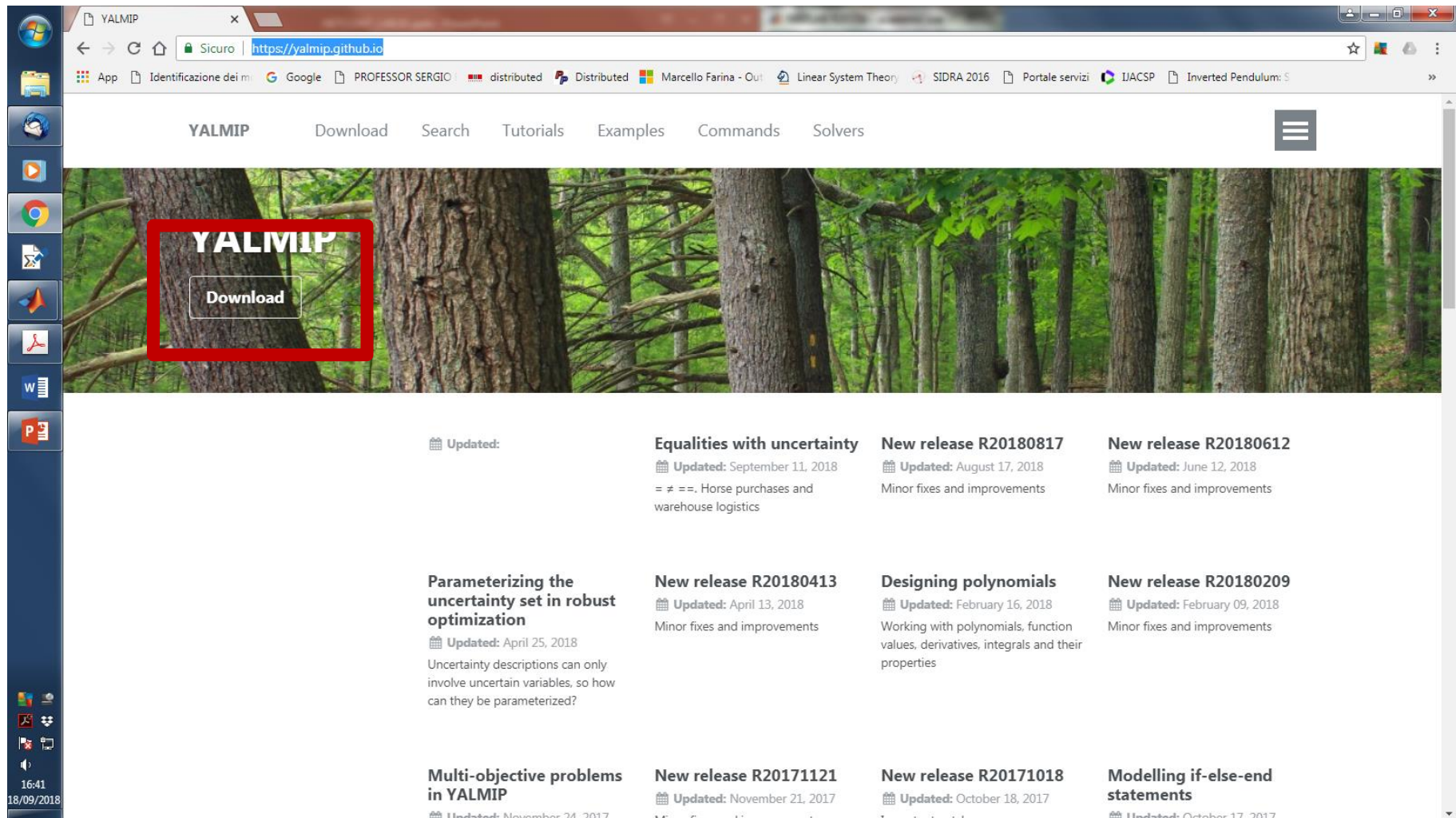
Politecnico di Milano, Italy

`marcello.farina@polimi.it`

TUESDAY, 28th SEPTEMBER, the first computer session will take place.

- BRING YOUR LAPTOP (IF YOU ARE IN THE CLASS), WITH MATLAB INSTALLED
- FOLLOW THE INSTRUCTIONS FOR ADDITIONAL SOFTWARE INSTALLATION

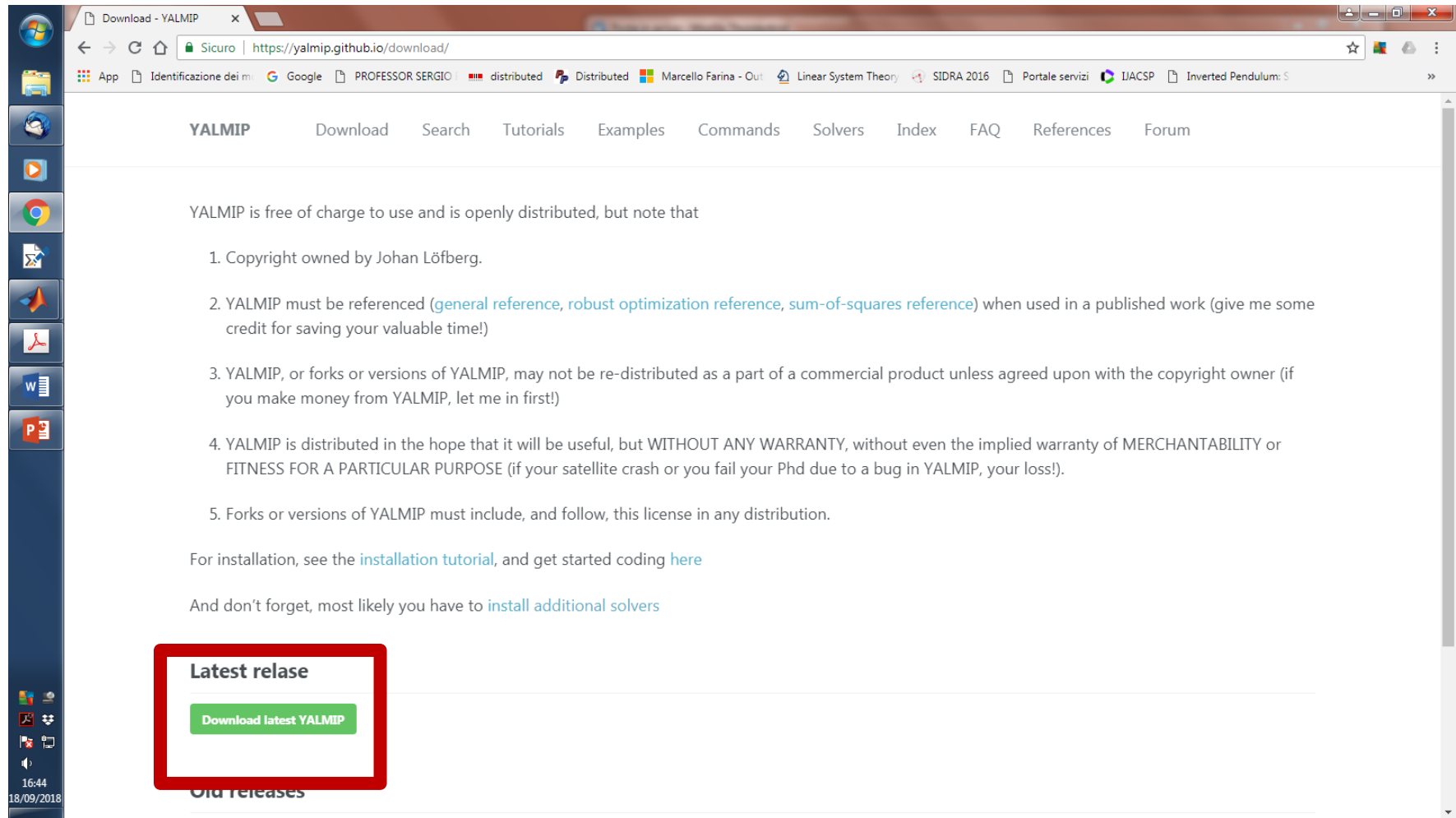
1. Download YALMIP at <https://yalmip.github.io/>



The screenshot shows a web browser window with the URL <https://yalmip.github.io/>. The page features a navigation bar with links: YALMIP, Download, Search, Tutorials, Examples, Commands, and Solvers. A large banner image of a forest is displayed, with the word "YALMIP" and a "Download" button highlighted by a red rectangular box. Below the banner, there are several sections for updates and releases, each with a title, an update date, and a brief description.

Update Title	Update Date	Description
Equalities with uncertainty	Updated: September 11, 2018	$= \neq =$, Horse purchases and warehouse logistics
New release R20180817	Updated: August 17, 2018	Minor fixes and improvements
New release R20180612	Updated: June 12, 2018	Minor fixes and improvements
Parameterizing the uncertainty set in robust optimization	Updated: April 25, 2018	Uncertainty descriptions can only involve uncertain variables, so how can they be parameterized?
New release R20180413	Updated: April 13, 2018	Minor fixes and improvements
Designing polynomials	Updated: February 16, 2018	Working with polynomials, function values, derivatives, integrals and their properties
New release R20180209	Updated: February 09, 2018	Minor fixes and improvements
Multi-objective problems in YALMIP	Updated: November 24, 2017	
New release R20171121	Updated: November 21, 2017	Minor fixes and improvements
New release R20171018	Updated: October 18, 2017	Important patch
Modelling if-else-end statements	Updated: October 17, 2017	

1. Download YALMIP at <https://yalmip.github.io/>



The screenshot shows a web browser window with the address bar displaying <https://yalmip.github.io/download/>. The page has a navigation bar with links: YALMIP, Download, Search, Tutorials, Examples, Commands, Solvers, Index, FAQ, References, and Forum. The main content area states that YALMIP is free of charge and openly distributed, followed by five numbered points regarding copyright, referencing, redistribution, warranty, and license inclusion. Below this, there are links for installation and solvers. At the bottom, a red-bordered box highlights the 'Latest release' section, which contains a green button labeled 'Download latest YALMIP'. The 'Old releases' section is partially visible below.

Download - YALMIP x

Sicuro | <https://yalmip.github.io/download/>

App Identificazione dei m Google PROFESSOR SERGIO distributed Distributed Marcello Farina - Out Linear System Theory SIDRA 2016 Portale servizi IJACSP Inverted Pendulum: S

YALMIP Download Search Tutorials Examples Commands Solvers Index FAQ References Forum

YALMIP is free of charge to use and is openly distributed, but note that

1. Copyright owned by Johan Löfberg.
2. YALMIP must be referenced ([general reference](#), [robust optimization reference](#), [sum-of-squares reference](#)) when used in a published work (give me some credit for saving your valuable time!)
3. YALMIP, or forks or versions of YALMIP, may not be re-distributed as a part of a commercial product unless agreed upon with the copyright owner (if you make money from YALMIP, let me in first!)
4. YALMIP is distributed in the hope that it will be useful, but WITHOUT ANY WARRANTY, without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE (if your satellite crash or you fail your Phd due to a bug in YALMIP, your loss!).
5. Forks or versions of YALMIP must include, and follow, this license in any distribution.

For installation, see the [installation tutorial](#), and get started coding [here](#)

And don't forget, most likely you have to [install additional solvers](#)

Latest release

Download latest YALMIP

Old releases

2. Download sedumi solver at <https://github.com/SQLP/SeDuMi>

The screenshot shows the GitHub repository page for `sqlp / sedumi`. The page includes a navigation bar with links to Features, Platform, Business, Explore, and Pricing. Below the navigation bar, the repository name `sqlp / sedumi` is displayed, along with statistics: 20 Watch, 101 Star, and 53 Fork. The main content area shows a banner for "Join GitHub today" and a section for the repository details. The repository is described as "SeDuMi: A linear/quadratic/semidefinite solver for Matlab and Octave" with a link to <http://coral.ie.lehigh.edu/~newsedumi/>. The repository has 39 commits, 2 branches, 0 releases, and 3 contributors. A red box highlights the "Clone or download" button and the "GPL-2.0" license information. Below this, a list of files is shown, including `conversion`, `doc`, `examples`, `o_win`, and `.gitignore`, each with a brief description and the time since the last commit.

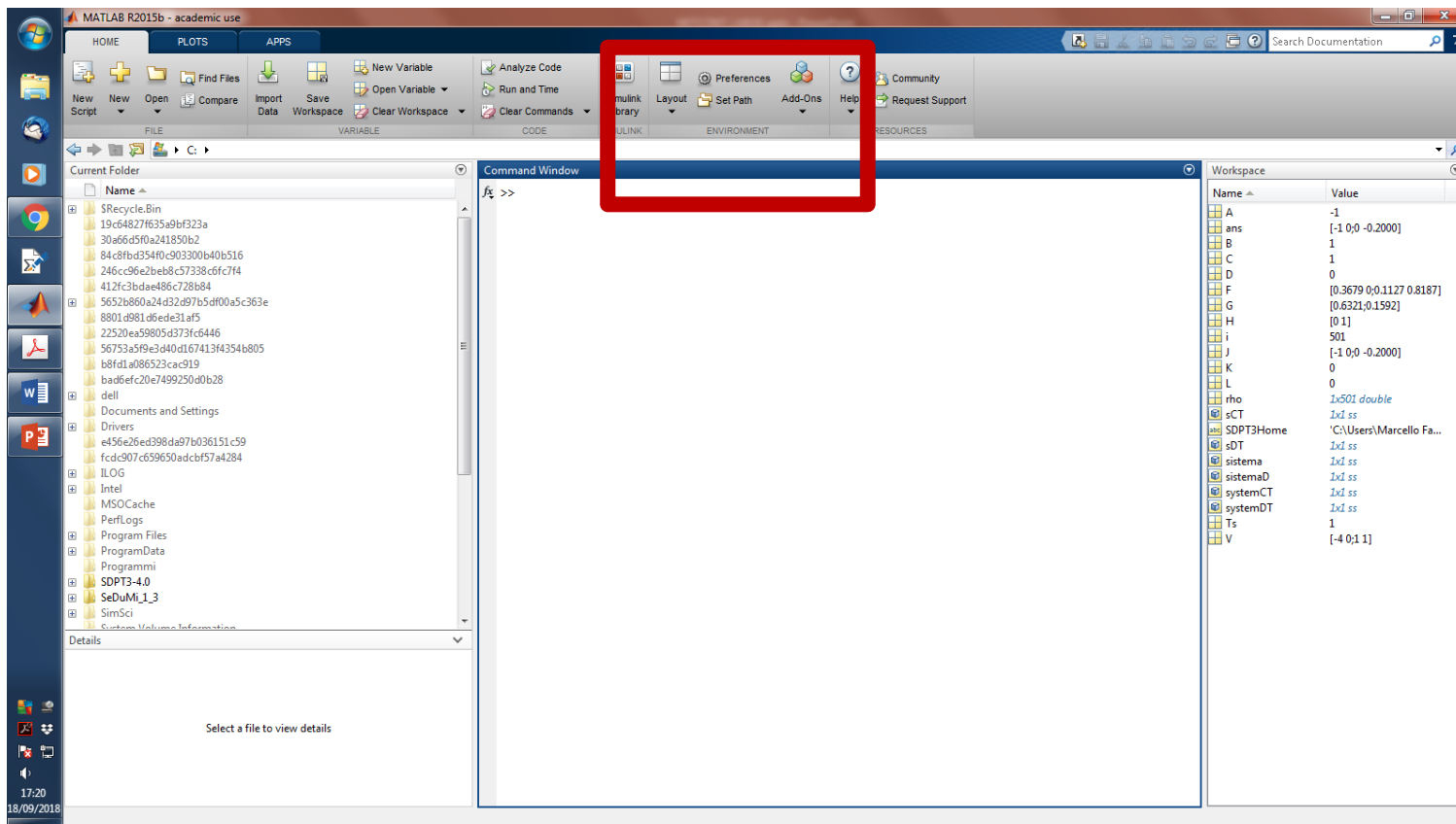
File	Description	Time since last commit
<code>conversion</code>	Lots of minor edits to suppress matlab code analyzer messages. Functi...	5 years ago
<code>doc</code>	Initial commit	6 years ago
<code>examples</code>	Initial commit	6 years ago
<code>o_win</code>	Added missing Octave MEX files	3 years ago
<code>.gitignore</code>	Added default .gitignore file	6 years ago

3. Download SDTP3 solver at <https://github.com/SQLP/SDPT3>

The screenshot shows the GitHub repository page for `sqlp / sdpt3`. The page includes a navigation bar with links to Features, Platform, Business, Explore, and Pricing. Below the repository name, there are statistics: 26 commits, 2 branches, 0 releases, and 2 contributors. A red box highlights the '2 contributors' section and the 'Clone or download' button. Below this, there is a list of files and their commit history:

File	Commit Message	Time
Examples	Initial commit	5 years ago
HSDSolver	Octave 4.0 changes; Windows Octave 4.0 binaries created	3 years ago
Solver	Recompile for mac; remove mexmaci	9 months ago
dimacs	Initial commit	5 years ago
sdplib	Initial commit	5 years ago

4. Unzip Yalmip, sedumi, and SDTP3 folders in an appropriate position
5. Open Matlab
5. Add the three unzipped folders (with subfolders) to the Matlab path



4. Unzip Yalmip, sedumi, and SDTP3 folders in appropriate positions
5. Open Matlab
6. Add the three unzipped folders (with subfolders) to the Matlab path

